

HEALTHCARE MONITORING SYSTEM

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HEALTHCARE MONITORING SYSTEM USING IoT

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ABSTRACT

Healthcare Monitoring System is an IoT-based smart healthcare solution designed to continuously monitor a patient's vital parameters such as body temperature, heart rate, and blood oxygen level. The system uses sensors connected to an ESP32 microcontroller for data acquisition. The collected data is transmitted through Wi-Fi to a cloud platform where it can be monitored remotely by doctors, caregivers, and family members. This system provides real-time health monitoring, early detection of abnormalities, and improved patient care. The project demonstrates the application of IoT technology in modern healthcare systems.

Keywords: IoT, Healthcare Monitoring, ESP32, Heart Rate Sensor, Temperature Sensor, Cloud Monitor

INTRODUCTION

Healthcare monitoring is essential for tracking patient health conditions. Traditional monitoring methods require manual observation and hospital visits. IoT technology enables continuous monitoring of vital signs and remote access to patient data.

Objectives

- Monitor body temperature.
- Measure heart rate.
- Measure blood oxygen saturation (SpO₂).
- Upload health data to cloud platforms.
- Provide remote monitoring through mobile devices.
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LITERATURE SURVEY

Several healthcare monitoring systems have been developed using wireless communication technologies. Existing systems mainly focus on hospital-based monitoring. Recent IoT solutions provide real-time data collection and cloud integration, improving healthcare accessibility and efficiency.

SYSTEM ANALYSIS Existing System

- Manual health monitoring.
- Requires hospital visits.
- No continuous monitoring.
- Delayed emergency response.

Proposed System

- Real-time monitoring.
- Cloud-based data storage.
- Remote access through mobile devices.
- Early health issue detection.

SYSTEM DESIGN Block Diagram

Patient → Sensors → ESP32 Controller → Wi-Fi → Cloud Server → Mobile App/User

Components

1. ESP32 Development Board
2. Pulse Sensor
3. MAX30102 Sensor
4. LM35/DS18B20 Temperature Sensor
5. Wi-Fi Module (Built into ESP32)
6. Power Supply

HARDWARE REQUIREMENTS

ESP32

ESP32 is a low-power microcontroller with built-in Wi-Fi and Bluetooth capabilities.

Pulse Sensor

Measures the heart rate of the patient.

MAX30102

Measures blood oxygen saturation and pulse rate.

Temperature Sensor

Measures body temperature continuously.

Power Supply

Provides power to the system components.

SOFTWARE REQUIREMENTS Software Purpose Arduino IDE Programming ESP32
Embedded C Firmware Development Blynk IoT Platform Cloud Monitoring Thing Speak
Data Visualization

METHODOLOGY Step 1

Sensors collect body temperature, heart rate, and oxygen level.

Step 2

ESP32 processes the sensor data.

Step 3

Data is transmitted through Wi-Fi.

Step 4

Cloud platform stores and displays data.

Step 5

Users access data through mobile application.

CIRCUIT DIAGRAM Connections

LM35 → ESP32 Analog Pin

Pulse Sensor → ESP32 GPIO Pin

MAX30102 → I2C Interface (SDA, SCL)

ESP32 → Wi-Fi Network

Power Supply → Entire System

IMPLEMENTATION

The system is programmed using Arduino IDE. Sensors continuously collect health data. ESP32 reads the values and sends them to the cloud using Wi-Fi. The data is displayed in real time on a dashboard.

RESULTS AND DISCUSSION

The developed healthcare monitoring system successfully measured:

- Body Temperature
- Heart Rate
- Blood Oxygen Level

The data was uploaded to the cloud and monitored remotely using a smartphone application.

Sample Results

Temperature: 36.8°C

Heart Rate: 78 BPM

SpO₂: 98%

The system achieved accurate and reliable monitoring.

ADVANTAGES

- Low cost
 - Portable
 - Real-time monitoring
 - Remote healthcare support
 - Easy installation
 - Reduced hospital visits
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APPLICATIONS

- Hospitals
- Home healthcare
- Elderly patient monitoring
- Rural healthcare centers
- Emergency monitoring systems

FUTURE SCOPE

- AI-based disease prediction.
- SMS emergency alert system.
- Integration with wearable devices.
- GPS-based patient tracking.
- Machine learning for health analytics.

CONCLUSION

The Healthcare Monitoring System using IoT successfully monitors vital health parameters and uploads data to cloud servers for remote access. The system improves patient care by providing continuous monitoring and timely medical intervention. It is cost-effective, reliable, and suitable for modern healthcare applications.

REFERENCES

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